

$i, j, k, l, m, n$

$x$

$t, \text{ term} ::=$   
 $\quad | \quad x$   
 $\quad | \quad \textbf{letrec } lb_1 \textbf{ and ... and } lb_n \textbf{ in } t$

$lb, \text{ letrec\_binding} ::=$   
 $\quad | \quad x = t$

$b, \text{ binding} ::=$   
 $\quad | \quad x$

$E, \text{ environment} ::=$   
 $\quad | \quad b_1, \dots, b_n$   
 $\quad | \quad \textbf{nil} \quad \quad \quad \text{M}$   
 $\quad | \quad E_1, \dots, E_n \quad \quad \quad \text{M}$

$\text{formula} ::=$   
 $\quad | \quad \text{judgement}$

$\text{Typing} ::=$   
 $\quad | \quad E| - \textbf{ok}$   
 $\quad | \quad E| - t$   
 $\quad | \quad E| - lb \textbf{ gives } E'$

$\text{judgement} ::=$   
 $\quad | \quad \text{Typing}$

$\text{user\_syntax} ::=$   
 $\quad | \quad i$   
 $\quad | \quad x$   
 $\quad | \quad t$   
 $\quad | \quad lb$   
 $\quad | \quad b$   
 $\quad | \quad E$

$E| - \textbf{ok}$

$\frac{}{\textbf{nil}| - \textbf{ok}} \quad \text{TYPING\_OK\_NIL}$

$\frac{E| - \textbf{ok}}{E, x| - \textbf{ok}} \quad \text{TYPING\_OK\_X}$

$E| - t$

$\frac{E_1, x, E_2| - \textbf{ok}}{E_1, x, E_2| - x} \quad \text{TYPING\_TERM\_X}$

<<no parses (char 35):  $E, E_1, \dots, E_n | - lb_1 \textbf{ gives } E_1 \textbf{ .***. } E, E_1, \dots, E_n | - lb_n$  >>

$E, E_1, \dots, E_n| - t$

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$E| - \textbf{letrec } lb_1 \textbf{ and ... and } lb_n \textbf{ in } t$

$$\boxed{E| - lb \textbf{gives} E'}$$

$$\frac{E| - t}{E| - x = t \textbf{gives} x} \quad \text{TYPING\_LB\_SIMPLE}$$

Definition rules: 4 good 1 bad

Definition rule clauses: 9 good 1 bad